THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of the first 15 odd natural numbers is: 1

(a) 225 **(b)** 8 **(c)** 16 **(d)** 15

2. A data set 58, 33, 23, 18, 39, 59, 53 gets an extra (outlier) value 580. The median of the new data of 8 values is: 1

(a) 107.88 **(b)** 46 **(c)** 580 **(d)** 39

3. A car travels 90 km at 30 km/h and then 90 km at 45 km/h. Its average speed (km/h) for the whole journey is: 1

(a) 65 **(b)** 36 **(c)** 37.5 **(d)** 18

4. The mean of 10 numbers is 22. If every number is multiplied by 2 and then 7 is added, the new mean is: 1

(a) 29 **(b)** 44 **(c)** 58 **(d)** 51

5. The mean of 5 numbers is 45. When one number is removed the mean of the remaining 4 becomes 40. The removed number is: 1

(a) 20 **(b)** 5 **(c)** 65 **(d)** 45

6. The mean of 15 values was found to be 60. Later it was found that 75 was misread as 45. The correct mean is: 1

(a) 90 **(b)** 58 **(c)** 60 **(d)** 62

7. A class has 34 students with mean age 14 years. When the teacher's age is included, the mean rises to 15. The teacher's age (years) is: 1

(a) 48 **(b)** 15 **(c)** 44 **(d)** 49

8. The mean of the first 21 odd natural numbers is: 1

(a) 22 **(b)** 11 **(c)** 21 **(d)** 441

9. **Assertion (A):** The mode of 2, 3, 3, 5, 5, 5, 7 is 5. 1

Reason (R): The mode is the middle value of the ordered data.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The median of the four observations 3, 5, 8, 10 is 8. 1
Reason (R): For an even number of observations, the median is the mean of the two middle values.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The mean of the first 11 even natural numbers is 12. Verify this for n = 11. 2

12. A data set was recorded as 38, 50, 23, 42, 20, 43, but the first value should have been 32. Find the correct mean. 2

13. The mean of 6 numbers is 14. When one number is removed, the mean of the remaining 5 numbers is 12. Find the removed number. 2

14. The mean of the first 14 even natural numbers is 15. Verify this for n = 14. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A class of 30 students has mean mark 55. After rechecking, a score of 36 was corrected to 66. Find the new mean. 3

16. The mean age of 14 players is 25 years. When the coach is included the mean becomes 27. Find the age of the coach. 3

17. A student's marks in 5 tests are 54, 59, 68, 64, 60. What mark must she score in the 6th test so that her mean becomes 67? 3

OR

A class of 20 students has mean mark 72. After rechecking, a score of 52 was corrected to 92. Find the new mean. 3

18. The mean age of 9 players is 30 years. When the coach is included the mean becomes 32. Find the age of the coach. 3

OR

A student's marks in 5 tests are 66, 57, 75, 53, 59. What mark must she score in the 6th test so that her mean becomes 68? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A dot plot shows the number of siblings of 10 students in a class: two students have 0 siblings, four have 1, three have 2 and one has 3. 5
(i) How many students are there in all?
(ii) What is the mode of the data?
(iii) How many students have at least 2 siblings?
(iv) Find the mean and the median number of siblings.

20. The mean of 20 numbers is 30, the mean of 10 numbers is 25 and the mean of 20 numbers is 70. Find the mean of all 50 numbers. 5

OR

The mean of 10 observations is 30. If each observation is multiplied by 2 and then decreased by 11, find the new mean and the new sum. 5

END OF QUESTION PAPER — Total: 40 Marks